# Exploring Inhalation Toxicity of Chemical Compounds

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05/13/2025

## 1 Introduction

Humans are constantly exposed to a diverse array of synthetic and naturally occurring chemical compounds in their environment. Many of these substances are incorporated into everyday consumer products, including fragrances. Therefore, it is critical to determine whether the chemical compounds used in fragrance formulations are truly safe for human health, particularly with respect to inhalation exposure.

Our study on fragrance toxicity involved three datasets: one containing chemical compounds known to be toxic when inhaled, another consisting of fragrance compounds with unknown toxicity, and a third matching the unknown fragrance compounds with an associated toxicity level-low (Class I), intermediate (Class II), and high (Class III). In the third dataset, toxicity outcomes were obtained using the Toxtree module's Revised Cramer Rules [1], a tool available for free download and widely used in toxicology for predicting toxicity based on chemical structure.

In our analysis, compounds classified as Class II or III were labeled as toxic, while Class I compounds were considered safe. We sourced these datasets from PubChem, a chemical database maintained by the National Institutes of Health (NIH) [2], and the Environmental Protection Agency (EPA) [3]. Initially, we obtained a list of chemicals from PubChem and then cross-referenced it with the EPA database to gather a comprehensive dataset that includes the relevant variables.

The dataset of toxic compounds totals roughly 1500, while the unknown fragrance and toxtree datasets total roughly 3000. After some cleaning in Excel, the unknown fragrance dataset observations were reduced to about 1600. In our study, we shuffled the unknown and toxtree data to create a merged mixed toxic and non-toxic compound dataset for regression analysis.

We have two central questions regarding our datasets: 1. Are Ames Test Predictions significant in identifying similarities between toxic compounds? 2. What variables regarding toxic and non-toxic chemical compounds are the strongest predictors for inhalation toxicity?

For our first modeling method, we employed clustering to explore the toxic inhalation dataset in relation to Ames Test Predictions. In chemistry, the Ames Test is used to discern the capacity for a chemical to trigger cell mutations (mutagenesis), which may lead to cancer in humans. A positive Ames test value indicates higher mutagenicity, while a negative test value indicates no genotoxicity. We chose to use this variable as it was present in the toxic dataset and not the unknown, so we wished to understand if Ames Test results were at all meaningful in finding similarities between toxic compounds.

For our second modeling method, the variables we chose for further study through regression in the fragrance compound and toxic inhalation datasets included vapor pressure, water solubility, and logP, as these properties are correlated with compound inhalation toxicity and were shared variables in both datasets. We aimed to identify which of these predictors most strongly influence inhalation toxicity, with the potential to refine the model and incorporate other relevant predictors in future work.

## 2 Data Processing

While reading in all datasets, we used the na.strings function to identify rows containing N/A in the datasets and convert them to NA's understood by R. Next, we removed extraneous columns irrelevant to our study in the toxic dataset and changed categorical columns (such as names of compounds and compound identifiers) to factors in all datasets. After, we performed a right-join merge on the toxtree data and unknown fragrance compound dataset to create a new known dataset, while removing duplicate values. This reduced the new dataset observations to 63. Then, we removed NA values and extreme outliers for some variables in the toxic dataset, though as we are dealing with chemical compounds a wide variance in values still remained. Removing outliers and missing values for the toxic dataset reduced the total observations to 387. Finally, we performed an outer-join merge on the toxic dataset and the new, mixed dataset, removing duplicate rows shared between the two to use later in our regression model. This merged dataset has 422 observations.

## 3 Data Exploration

First, we analyzed the distribution of Ames test predictions in the toxic dataset. The histogram plot and summary table below for the Ames Mutagenicity Test Predictions show that the data is right-skewed, with the largest portion of observations having a test prediction between 0-0.2. This indicates that the majority of observations in the toxic chemicals dataset have slight mutagenetic properties.

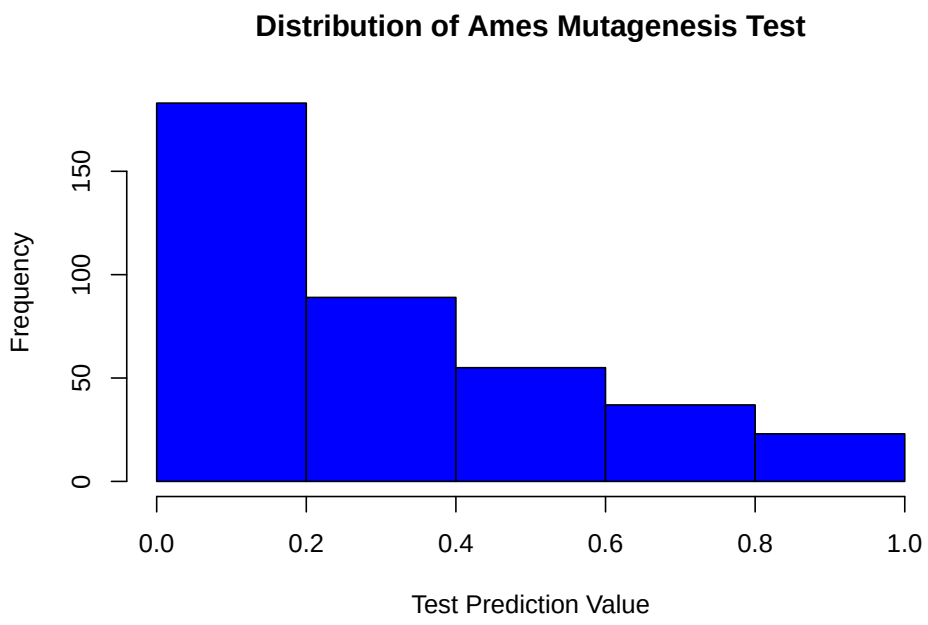



Figure 1: Distribution of Ames Mutagenesis Test Plot

Table 1: Descriptive Statistics for Ames Mutagenicity Test Predictions:

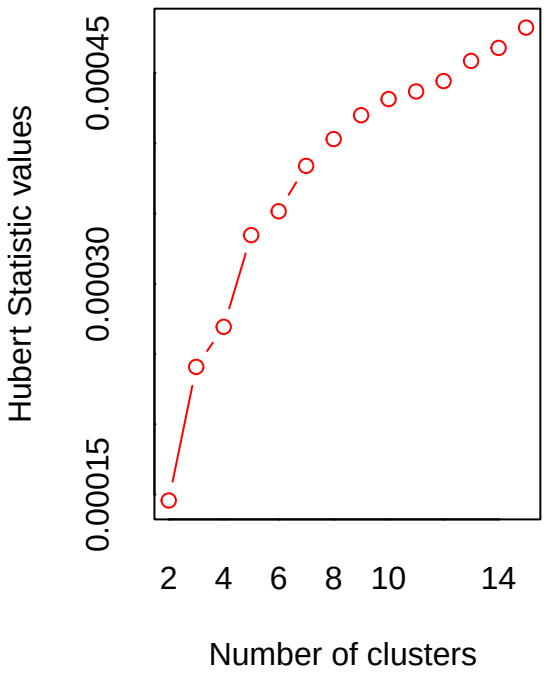
| Min | Max | Median | Mean |
|---|---|---|---|
| 0.001 | 0.995 | 0.214 | 0.2917984 |

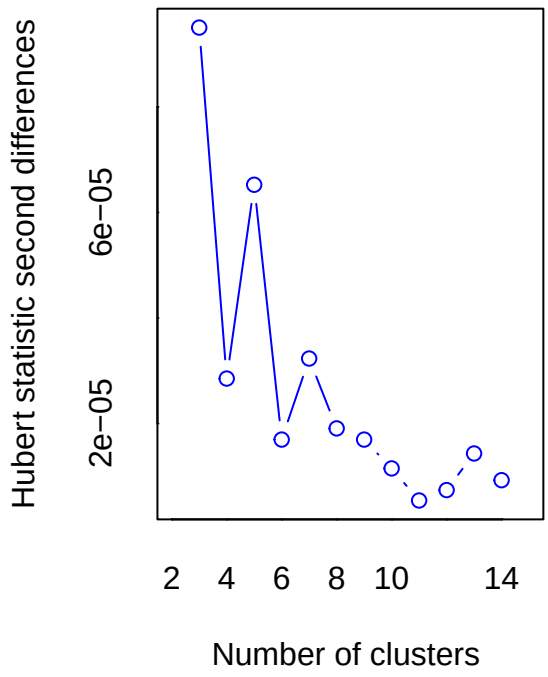
# 4 Method 1: Clustering

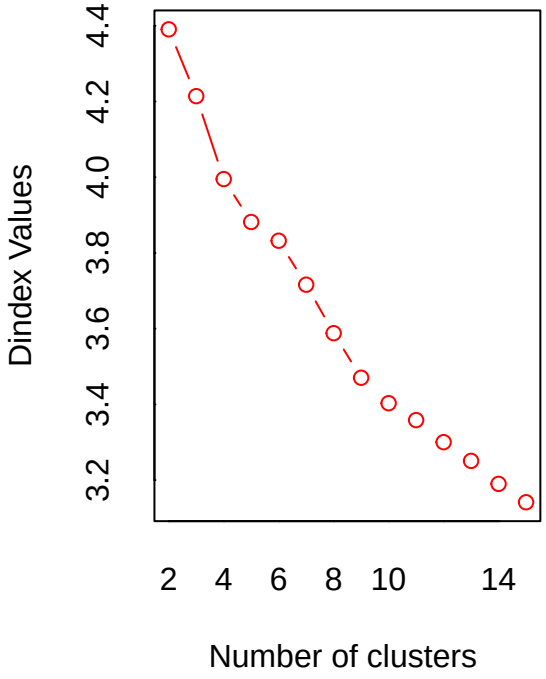
In this section, we aim to discover more about the toxic dataset through two clustering methods: K-Means and Gaussian Mixture. We hope to learn if there are any meaningful groupings in our toxic dataset, and if these groupings are at all correlated with Ames Test Prediction values.

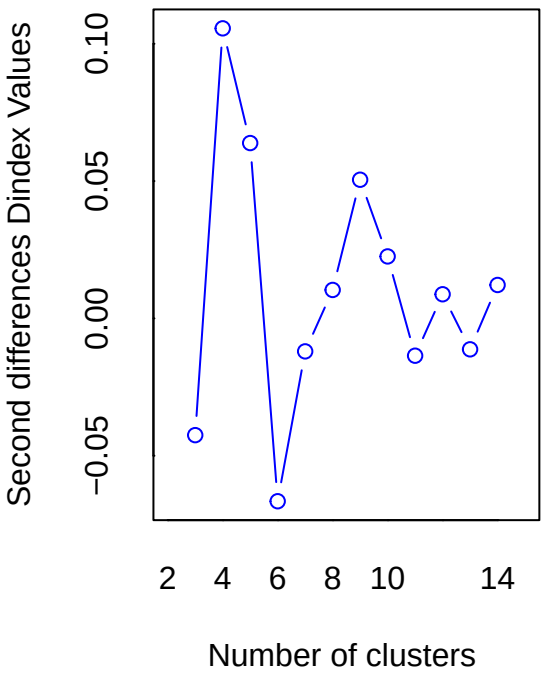
# 5 Training and Testing Data Sets

To create proper clusters, we first removed all categorical variables from the toxic dataset, assigning the result to a new dataframe, inhalation2.data. Then, we scaled the results as there is wide variety between predictor values. To determine ideal parameters for clustering, used the NbClust package to automate the ideal kmeans cluster amount. The results below indicate an efficient cluster number would be around 6-8, where the graph comes to a "point," or "elbow."




Dindex Values
3.2
3.4
3.6
3.8
4.0
4.2
4.4
2
4
6
8
10
14
Number of clusters


Second differences Dindex Values
−0.05
0.00
0.05
0.10
2
4
6
8
10
14
Number of clusters

# 6 Modeling

In this section, we employed two different clustering methods- K-Means clustering, and Gaussian Mixture Model-Based clustering. Our K-Means model is fed a pre-determined amount of clusters, or groups, and identifies data points' locations based on the value of each cluster's center, or centroid. As for Gaussian Mixture clustering, the model also aims to label data as being part of a particular cluster, but does this using probability distributions as opposed to means. As for a variable of interest, we chose the Ames Mutagenicity test for its smaller distribution compared to several other potential toxicity predictors with wider variance.

## 6.1 K-Means Clustering

To fit a k-means cluster model, we passed in the scaled inhalation data, instructing the model to use 7 clusters. We decided to use a low nstart value of 5 due to the large variance in the dataset even after scaling, which distributed the data between clusters more meaningfully.

### 6.1.1 K-Means Model Fit Results

The size of each cluster varies widely, with the smallest cluster having only 10 values, and the largest cluster with 126. The measure of distance within clusters is greater than the measure of distance between clusters, meaning there is likely no significant trend among values in the dataset as a whole.

Table 2: Size Of Each Cluster

| Cluster Amount | Cluster |
|---|---|
| 10 | 1 |
| 11 | 1 |
| 19 | 1 |
| 61 | 1 |
| 69 | 1 |
| 91 | 1 |
| 126 | 1 |

Table 3: Measure of Within Distance Of Each Cluster

| Measure | Freq |
|---|---|
| 6748.92906918926 | 1 |

Table 4: Measure of Between Distance Of Each Cluster

| Measure | Freq |
|---|---|
| 5217.07093081074 | 1 |

### 6.1.2 K-Means Clustering Prediction Performance

We then analyzed the fit.inhalation model in terms of one of our variables of interest, Ames Mutagenicity Test Prediction. The table below analyzes the scaled means for each cluster, while the plot illustrates the clustering model itself. The table and plot appear to reflect the data above- there are certainly enough differences between values as a whole to create clusters, but there is still greater variety within clusters than between them. This may indicate the data itself is generally similar- as these are all toxic compounds with some degree of mutagenicity.

Table 5: Means For Each Cluster of Ames Test

| Means | Cluster_Number |
|---|---|
| -0.2011762 | 1 |
| 0.0912903 | 2 |
| 0.4243295 | 3 |
| 0.0517261 | 4 |
| 0.1528265 | 5 |
| -0.5350234 | 6 |
| -0.5355170 | 7 |

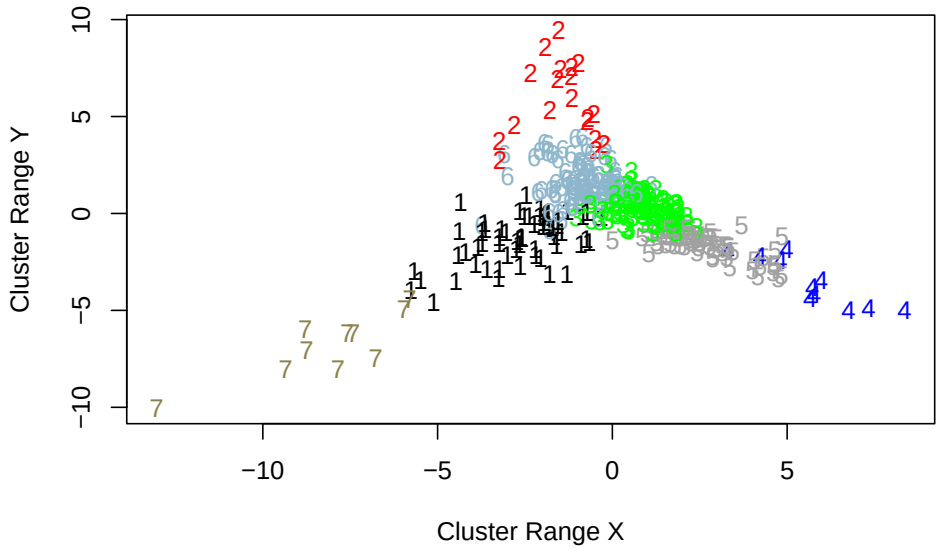



Figure 2: K-Means Clustering For Ames Test

## 6.2 Gaussian Mixture Model-Based Clustering- Mclust

To fit the Mclust Gaussian Mixture model, we passed in the scaled inhalation data with no other parameters, as this method uses a mixture of models.

### 6.2.1 Mclust Fit Results

The Mclust model compared 14 different fitted models, with VEV being the most accurate until about 6 clusters. After that point, the EEV model is more accurate in determining cluster belonging.

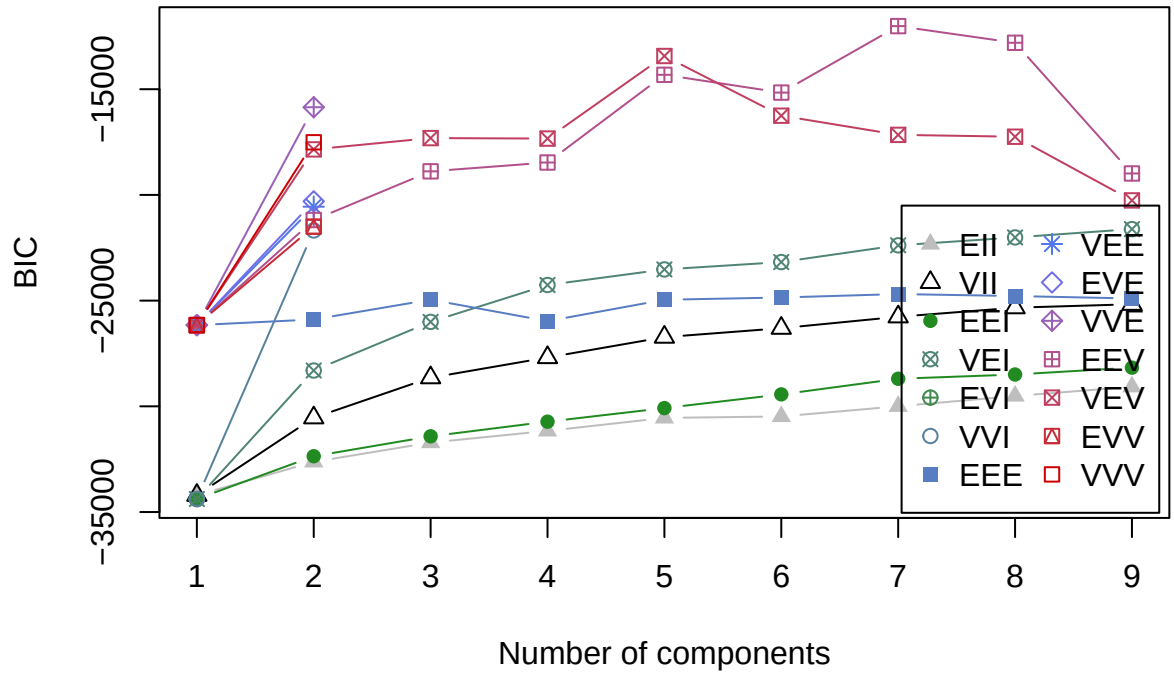



Figure 3: Mclust Inhalation Fit Model Parameters:

### 6.2.2 Mclust Clustering Prediction Performance

We then analyzed the inhalation.model.clus model in terms of our variable of interest, Ames Mutagenicity Test Prediction. We used the chi-square test to measure correlation between the chosen clusters and Ames Test Predictions. The returned p-value of 0.025 indicates there is a significant amount of correlation between Ames Test Pred values of data points and their associated clusters.

Table 6: Pearson's Chi-squared Test:

| Parameter | Data_Value |
|---|---|
| X-Squared: | 1801.48615532607 |
| df: | 1686 |
| p-value: | 0.0252020981327255 |
| Data Name: | table(inhalation.model.clus$classification, inhalation2.df$AMES_MUTAGENICITY_TEST_PRED) |

The plot below is a cleaner visual representation using dimension reduction of the ideal cluster belonging returned by the Mclust model. We can see through both K-Means and Gaussian Mixture Modeling that most of the clusters themselves are relatively similar to one another, but there may be one significantly different grouping of data in the dataset, indicated by cluster 7 in the K-Means Test and the yellow squares in the dimensional modeling below. The chi-square test results with a significant p-value also show that the Ames Test Predictions column could be useful to further categorize and understand the chemical properties of the given compounds.

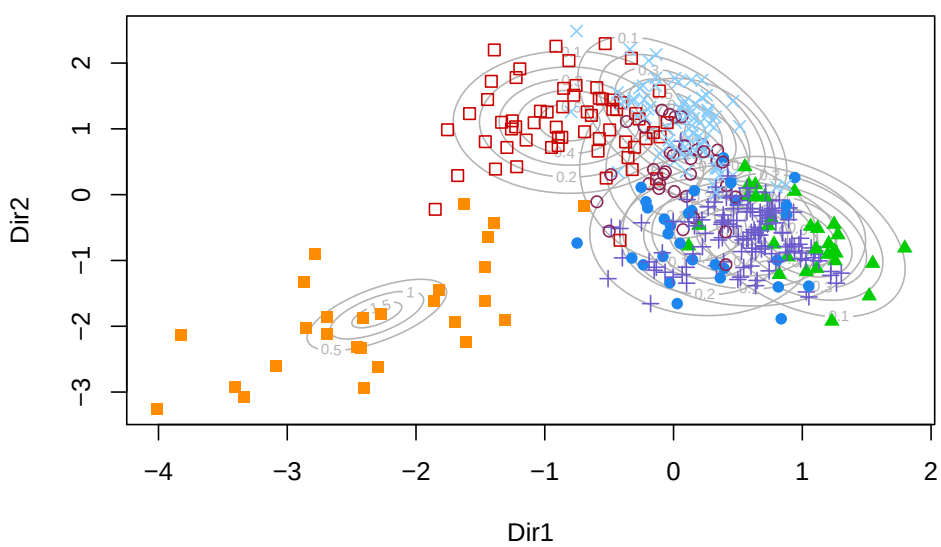



Figure 4: Dimensional Reduction Cluster Plot for Mclust Model

# 7 Method 2: Logistic Regression Classification Model

For our second model, we employ logistic regression. Using our merged dataset created earlier, and the toxicity variable, our goal is to predict a binary outcome: "toxic" or "non-toxic." We will evaluate the model's performance based on its accuracy and misclassification rate, and identify which predictors are most influential in determining inhalation toxicity. First, we analyzed the frequency of toxic and non-toxic observations in our merged dataset. As shown below in the barplot and frequency table, roughly ~11% of the data is in the minority class (safe), making our dataset moderately imbalanced.

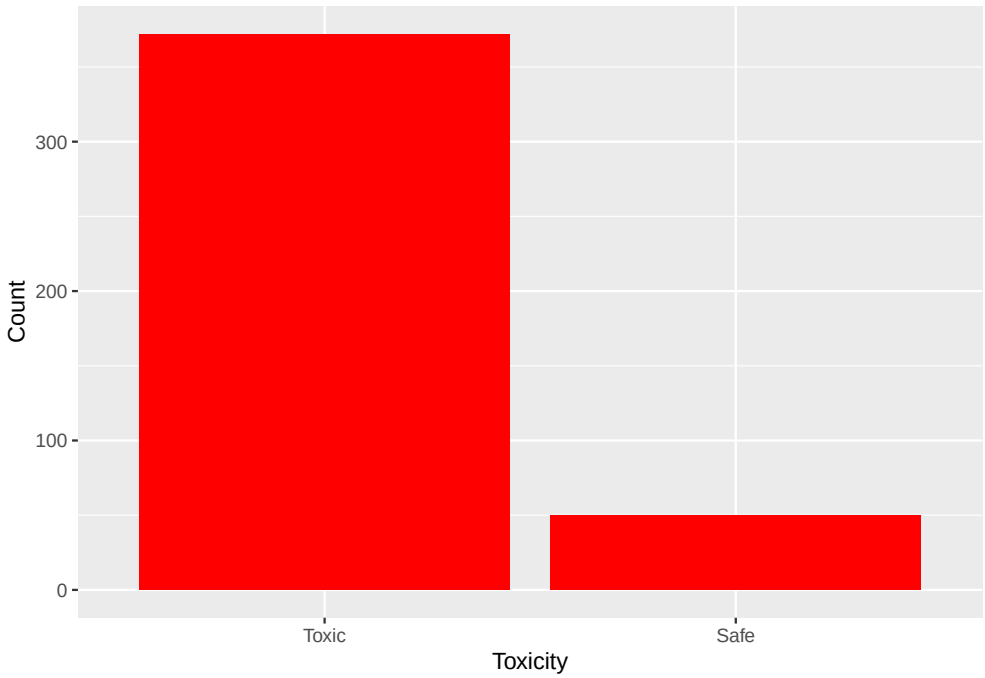



Figure 5: Frequency of Toxicity Plot

Table 7: Frequency Levels of Toxicity Observations

| Toxicity | Frequency |
|---|---|
| | 0 |
| Safe | 50 |
| Toxic | 372 |

# 8 Training and Testing Data Sets

We opted to shuffle our data points in the merged dataset, creating a new dataframe (shuffled_merge.df) before partitioning training and testing sets, as we merged both datasets together and there are far more toxic than non-toxic observations. This allows for the most balanced data possible in each set. We then created training and testing sets, using a standard 75% of data for the training set and the remaining 25% for testing. Finally, we used the ROSE function to further balance the training set before fitting a model.

# 9 Modeling

In this section, we fit a logistic regression model to our training and testing data to determine which predictors (XLog_P, Water Solubility, Vapor Pressure) were significant in identifying compound toxicity. Logistic regression is a method for determining a binary outcome using independent variables (predictors), and is an ideal model type as our outcome variable is binary in nature, either toxic or not.

## 9.1 Logistic Regression Model

We decided to fit the model to our three predictors of interest, XLog_P, Water Solubility, and Vapor Pressure. We used the glm method with ten cross-validation folds.

#### 9.1.1 Logistic Regression Model Fit Results

Our fitted model result table includes serveral different calculations. The "intercept" refers to the constant term in the dataset when all predictors are set to zero, while standard error and z values measure the variance of predicted outcomes using a predictor variable compared to the true value. The estimate parameter compares the projected change in our outcome, toxicity, if each predictor were to be increased by one unit. Finally, the p-value measures the statistical significance of each predictor value on the outcome, with values less than 0.05 deemed to be significant. In our case, the model determined that XLog_P is mildly significant in determining toxicity levels, while water solubility and vapor pressure were less important.
We then decided to plot an ROC curve, which measures prediction accuracy in a model. The larger the AUC (Area Under the Curve), the better the model. As shown below, our fitted toxicity model has an AUC measure of 0.57, which is only marginally better than a random model with a value of 0.50.

Table 8: Logistic Regression Model Predictor Results

| Coefficients | Estimate | Std_Error | Z_Value | P_Value |
|---|---|---|---|---|
| (Intercept) | 2.2605357 | 0.2959516 | 7.638 | 0.0000 |
| XLOG_P | -0.2059483 | 0.1008555 | -2.042 | 0.0412 |
| WATER_SOLUBILITY_MOL.L_TEST_PRED | 0.0421158 | 0.1862724 | 0.226 | 0.8211 |
| VAPOR_PRESSURE_MMHG_TEST_PRED | -0.0002361 | 0.0009452 | -0.250 | 0.8027 |

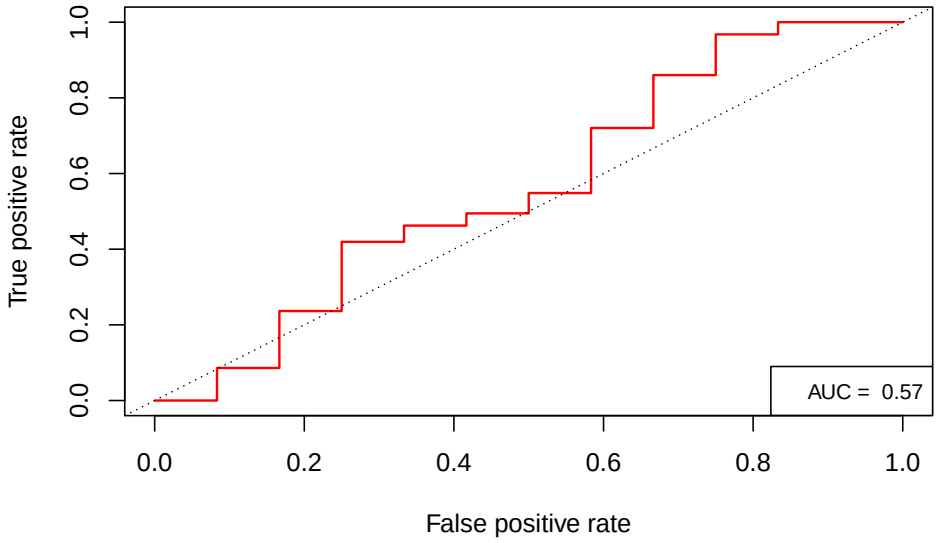



Figure 6: Toxicity Model Accuracy

### 9.1.2 Logistic Regression Prediction Performance

To further understand the performance of our model, we created a confusion matrix table to compare true positive and true negative rates with false positive and false negative rates. The confusion matrix results in the table below indicate the model was able to correctly identify toxic compounds, but was unable to identify safe compounds, labeling them toxic as well. We believe this is due to the imbalanced nature of the dataset and hence a lack of non-toxic observations to train our model on.

We then created a plot of the sensitivity (the proportion of identified true positives out of all actual positive values) and the specificity (the proportion of identified true negatives out of all actual negative values). This plot identifies the best threshold of values for determining the outcome classification. In this case, the plot determined that the ideal threshold is about 0.89. However, due to the lack of available training data for "safe" predictions, we were unable to test our predictions at such a high threshold.

Table 9: Predicted versus Actual Compound Toxicity Results

| Reference | Safe_Prediction | Toxic_Prediction |
|---|---|---|
| Safe | 0 | 0 |
| Toxic | 12 | 93 |

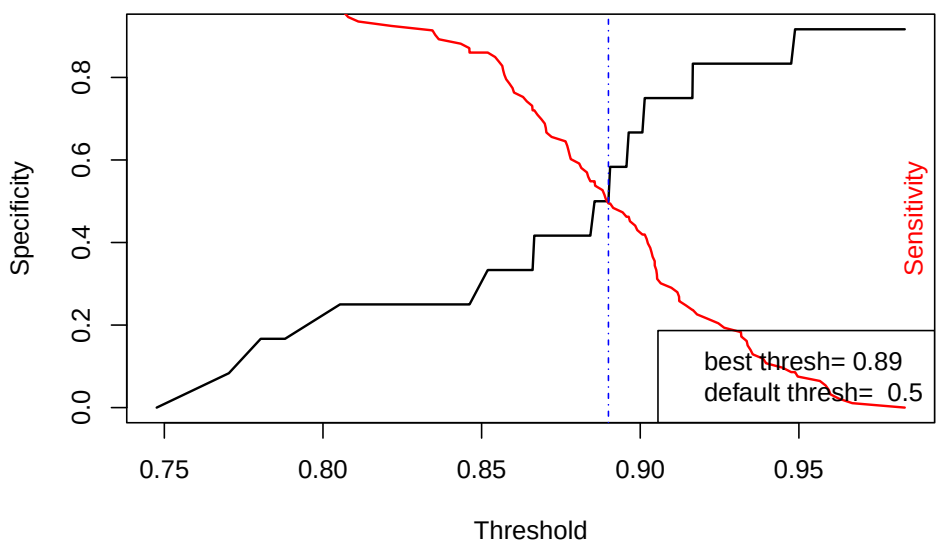



Figure 7: Sensitivity/Specificity Plot for Toxicity Levels in The Dataset

# 10 Conclusions

To address our first research question—whether Ames Test predictions are significant in identifying similarities between toxic compounds—we applied two clustering techniques: K-Means and Gaussian Mixture Models. While both methods produced at least one distinct cluster, most clusters did not show meaningful differentiation, and there was no strong separation based on mutagenicity. Although Ames Test predictions showed limited statistical significance, they did not align consistently with the overall cluster groupings. This suggests that Ames Test results are not strong indicators of structural or feature-based similarity among toxic compounds as captured by unsupervised clustering methods.

To explore our second research question—identifying the strongest predictors for inhalation toxicity—we applied logistic regression to a merged dataset of toxic and non-toxic compounds. Among the variables analyzed, XLog_P (a measure of lipophilicity) was the only statistically significant predictor. However, the model's predictive performance was only marginally better than random, likely due to class imbalance and the limited availability of complete non-toxic observations.

Key challenges included managing missing data, interpreting overlapping clusters, and selecting biologically meaningful features. Future work should address class imbalance by incorporating more complete non-toxic cases and explore additional predictor variables supported by toxicological literature to improve model robustness. Additionally, exploring alternative modeling strategies such as ensemble methods could move beyond binary classification and better capture the complex, continuous nature of toxicological effects.

# 11 References


1. Toxtree module's Revised Cramer Rules:
   https://toxtree.sourceforge.net/cramer.html

2. PubCHEM Datasets:
   https://pubchem.ncbi.nlm.nih.gov/classification/#hid=72&search=Fragrances
   https://pubchem.ncbi.nlm.nih.gov/classification/#hid=72&search=Inhalation+Risk

3. EPA Datasets:
   https://comptox.epa.gov/dashboard/batch-search